

Data Quality Across Relational Tables: Detecting Semantic Coherence Violations

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Can multi-table coherence be learned from raw data,
without manual rules or labeled errors ?

THE PROBLEM

PATIENT			PRESCRIPTION		
patient_id	age	ward	presc_id	id	dosage
P120	35	Cardiology	RX-87	P120	50 mg
P121	62	Oncology	RX-88	P121	75 mg
P123	4	Pediatrics	RX-89	P123	150 mg

patient_id → *id*

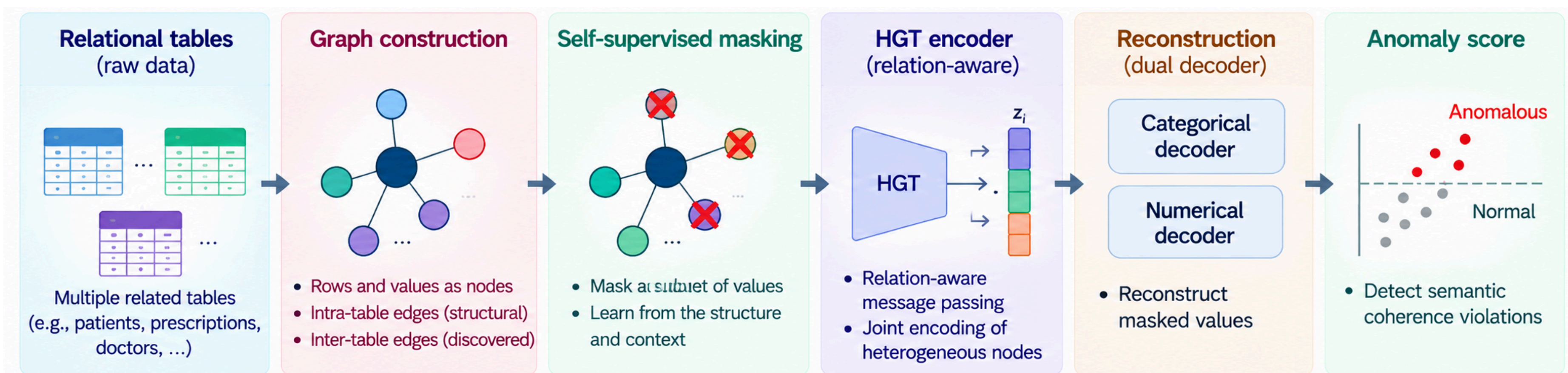


Every value is valid individually, but the combination is semantically incoherent.

Quality tools check each table independently. Cross-table violations remain invisible.

OUR APPROACH (GRACE)

A self-supervised GNN that learns relational coherence from raw data,
and detects semantic coherence violations across tables.



How can a graph model learn relational coherence across tables ?